

About the dataset

Dataset Description

The dataset contains historical and projected life expectancy data for countries around the world, spanning the years 1800 to 2100. Each record represents the average life expectancy for a specific country in a given year, enabling the analysis of long-term health trends across different regions.

The dataset includes key variables such as:

- name – Country name.
- year – The observation year (1800–2100).
- life_expectancy – Average life expectancy at birth, measured in years.

Overall, the dataset provides valuable insights into how advances in healthcare, nutrition, sanitation, education, and socioeconomic development have contributed to increasing life expectancy worldwide over time.

Data Source

In [1]:

```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
```

In [2]:

```
df = pd.read_csv("./data/life_expectancy.csv")
```

In [3]:

```
df.head()
```

Out[3]:

	geo	name	1800	1801	1802	1803	1804	1805	1806	1807	...	2091	2092	2093	2094	2095	2096	2097	2100
0	afg	Afghanistan	28.21	28.20	28.19	28.18	28.17	28.16	28.15	28.14	...	79.09	79.19	79.29	79.50	79.61	79.71	79.81	80.00
1	ago	Angola	26.98	26.98	26.98	26.98	26.98	26.98	26.98	26.98	...	73.56	73.66	73.76	73.95	74.05	74.15	74.35	74.50
2	alb	Albania	35.40	35.40	35.40	35.40	35.40	35.40	35.40	35.40	...	89.10	89.20	89.30	89.40	89.50	89.60	89.70	89.80
3	and	Andorra	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	91.89	92.09	92.19	92.29	92.39	92.49	92.59	92.70
4	are	UAE	30.70	30.70	30.70	30.70	30.70	30.70	30.70	30.70	...	87.64	87.73	87.83	87.93	88.02	88.12	88.22	88.30

5 rows x 303 columns



Understand the data

In [4]:

```
print(df.shape)
```

(194, 303)

In [5]:

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 194 entries, 0 to 193
Columns: 303 entries, geo to 2100
dtypes: float64(301), object(2)
memory usage: 459.4+ KB
None
```

In [6]:

```
print(df.isna().sum())
```

```
geo      0
name      0
1800     10
1801     10
1802     10
      ..
2096      1
2097      1
2098      1
2099      1
2100      1
Length: 303, dtype: int64
```

Clean the Data

Remove duplicates

In [7]:

```
df = df.drop_duplicates()
```

Check missing values

In [8]:

```
missing = (
    df.isna()
      .sum()
      .sort_values(ascending=False)
)

missing.head(20)
```

Out[8]:

```
1949     10
1893     10
1895     10
1896     10
1897     10
1898     10
1899     10
1900     10
1901     10
1902     10
1903     10
1904     10
1905     10
1906     10
1907     10
1908     10
1909     10
1894     10
1892     10
1912     10
dtype: int64
```

Convert from wide format to long format.

In [9]:

```
life = df.melt(
    id_vars=["geo", "name"],
    var_name="year",
    value_name="life_expectancy"
)
```

In [10]:

```
life.head()
```

Out[10]:

	geo	name	year	life_expectancy
0	afg	Afghanistan	1800	28.21
1	ago	Angola	1800	26.98
2	alb	Albania	1800	35.40
3	and	Andorra	1800	NaN
4	are	UAE	1800	30.70

In [11]:

```
print(life.shape)

(58394, 4)
```

Convert Year to numeric

In [12]:

```
life["year"] = life["year"].astype(int)
```

Convert Life Expectancy to numeric and drop the missing values

In [13]:

```
life["life_expectancy"] = pd.to_numeric(life["life_expectancy"], errors="coerce")
life = life.dropna(subset=["life_expectancy"]).reset_index(drop = True)
```

In [14]:

```
life.head()
```

Out[14]:

	geo	name	year	life_expectancy
0	afg	Afghanistan	1800	28.21
1	ago	Angola	1800	26.98
2	alb	Albania	1800	35.40
3	are	UAE	1800	30.70
4	arg	Argentina	1800	33.20

In [15]:

```
print(life['name'].unique())

['Afghanistan' 'Angola' 'Albania' 'UAE' 'Argentina' 'Armenia'
 'Antigua and Barbuda' 'Australia' 'Austria' 'Azerbaijan' 'Burundi'
 'Belgium' 'Benin' 'Burkina Faso' 'Bangladesh' 'Bulgaria' 'Bahrain'
 'Bahamas' 'Bosnia and Herzegovina' 'Belarus' 'Belize' 'Bolivia' 'Brazil'
 'Bhutan' 'Botswana' 'Bulgaria' 'Burkina Faso' 'Burundi'
 'Cameroon' 'Canada' 'Cape Verde' 'Central African Republic'
 'Chad' 'China' 'Colombia' 'Comoros' 'Congo' 'Costa Rica' 'Cote d'Ivoire'
 'Croatia' 'Cuba' 'Cyprus' 'Czechia' 'Democratic Republic of the Congo'
 'Denmark' 'Dominican Republic' 'Dominica' 'DRC' 'Ecuador' 'Egypt' 'El Salvador'
 'Equatorial Guinea' 'Estonia' 'Ethiopia' 'Fiji' 'Finland' 'France' 'Gabon'
 'Gambia' 'Germany' 'Ghana' 'Greece' 'Guatemala' 'Guinea' 'Guyana' 'Honduras'
 'Hungary' 'Iceland' 'India' 'Indonesia' 'Israel' 'Italy' 'Jamaica' 'Japan'
 'Jordan' 'Kazakhstan' 'Kenya' 'Kiribati' 'Kuwait' 'Kyrgyzstan' 'Laos' 'Latvia'
 'Lebanon' 'Lesotho' 'Liberia' 'Lithuania' 'Luxembourg' 'Madagascar' 'Malawi'
 'Malaysia' 'Maldives' 'Mali' 'Malta' 'Marshall Islands' 'Mauritania' 'Mauritius'
 'Mexico' 'Micronesia' 'Moldova' 'Mongolia' 'Montenegro' 'Morocco' 'Mozambique'
 'Myanmar' 'Namibia' 'Nauru' 'Nepal' 'Netherlands' 'New Zealand' 'Nicaragua'
 'Niger' 'Nigeria' 'North Macedonia' 'Norway' 'Oman' 'Pakistan' 'Palau' 'Panama'
 'Papua New Guinea' 'Paraguay' 'Peru' 'Philippines' 'Poland' 'Portugal' 'Puerto Rico'
 'Qatar' 'Romania' 'Russia' 'Rwanda' 'Saint Kitts and Nevis' 'Saint Lucia'
 'Saint Vincent and the Grenadines' 'Samoa' 'San Marino' 'Saudi Arabia' 'Senegal'
 'Serbia' 'Sierra Leone' 'Singapore' 'Slovakia' 'Slovenia' 'South Africa'
 'South Korea' 'South Sudan' 'Spain' 'Sri Lanka' 'Sudan' 'Suriname' 'Swaziland'
 'Sweden' 'Switzerland' 'Taiwan' 'Tajikistan' 'Tanzania' 'Thailand' 'Timor-Leste'
 'Togo' 'Tonga' 'Trinidad and Tobago' 'Tunisia' 'Turkey' 'Turkmenistan' 'Uganda'
 'Ukraine' 'United Kingdom' 'United States' 'Uruguay' 'Uzbekistan' 'Vanuatu'
 'Venezuela' 'Vietnam' 'Yemen' 'Zambia' 'Zimbabwe']
```

```
'Barbados' 'Brunei' 'Bhutan' 'Botswana' 'Central African Republic'
'Canada' 'Switzerland' 'Chile' 'China' 'Cote d'Ivoire' 'Cameroon'
'Congo, Dem. Rep.' 'Congo, Rep.' 'Colombia' 'Comoros' 'Cape Verde'
'Costa Rica' 'Cuba' 'Cyprus' 'Czech Republic' 'Germany' 'Djibouti'
'Denmark' 'Dominican Republic' 'Algeria' 'Ecuador' 'Egypt' 'Eritrea'
'Spain' 'Estonia' 'Ethiopia' 'Finland' 'Fiji' 'France'
'Micronesia, Fed. Sts.' 'Gabon' 'UK' 'Georgia' 'Ghana' 'Guinea' 'Gambia'
'Guinea-Bissau' 'Equatorial Guinea' 'Greece' 'Grenada' 'Guatemala'
'Guyana' 'Honduras' 'Croatia' 'Haiti' 'Hungary' 'Indonesia' 'India'
'Ireland' 'Iran' 'Iraq' 'Iceland' 'Israel' 'Italy' 'Jamaica' 'Jordan'
'Japan' 'Kazakhstan' 'Kenya' 'Kyrgyz Republic' 'Cambodia' 'Kiribati'
'South Korea' 'Kuwait' 'Lao' 'Lebanon' 'Liberia' 'Libya' 'St. Lucia'
'Sri Lanka' 'Lesotho' 'Lithuania' 'Luxembourg' 'Latvia' 'Morocco'
'Moldova' 'Madagascar' 'Maldives' 'Mexico' 'North Macedonia' 'Mali'
'Malta' 'Myanmar' 'Montenegro' 'Mongolia' 'Mozambique' 'Mauritania'
'Mauritius' 'Malawi' 'Malaysia' 'Namibia' 'Niger' 'Nigeria' 'Nicaragua'
'Netherlands' 'Norway' 'Nepal' 'New Zealand' 'Oman' 'Pakistan' 'Panama'
'Peru' 'Philippines' 'Papua New Guinea' 'Poland' 'North Korea' 'Portugal'
'Paraguay' 'Palestine' 'Qatar' 'Romania' 'Russia' 'Rwanda' 'Saudi Arabia'
'Sudan' 'Senegal' 'Singapore' 'Solomon Islands' 'Sierra Leone'
'El Salvador' 'Somalia' 'Serbia' 'South Sudan' 'Sao Tome and Principe'
'Suriname' 'Slovak Republic' 'Slovenia' 'Sweden' 'Eswatini' 'Seychelles'
'Syria' 'Chad' 'Togo' 'Thailand' 'Tajikistan' 'Turkmenistan'
'Timor-Leste' 'Tonga' 'Trinidad and Tobago' 'Tunisia' 'Turkey' 'Tanzania'
'Uganda' 'Ukraine' 'Uruguay' 'USA' 'Uzbekistan'
'St. Vincent and the Grenadines' 'Venezuela' 'Vietnam' 'Vanuatu' 'Samoa'
'Yemen' 'South Africa' 'Zambia' 'Zimbabwe' 'Andorra' 'Dominica'
'St. Kitts and Nevis' 'Liechtenstein' 'Monaco' 'Marshall Islands' 'Nauru'
'Palau' 'San Marino' 'Tuvalu']
```

Visualizations

In [20]:

```
def style_figure(fig, title, xlabel, ylabel, show_legend = True) -> go.Figure:
    """
    Applies custom styles to a Plotly figure to maintain a consistent theme:
    - Pure white background (paper and plot).
    - Light grey gridlines for readability without visual noise.
    - Highly readable dark-grey typography.
    """
    fig.update_layout(
        title={
            'text': f"<b>{title}</b>",
            'y': 0.95,
            'x': 0.05,
            'xanchor': 'left',
            'yanchor': 'top',
            'font': dict(size=18, color="#2C3E50")
        },
        paper_bgcolor='white',
        plot_bgcolor='white',
        xaxis=dict(
            title=f"<b>{xlabel}</b>",
            showgrid=True,
            gridcolor='#EAEAEA',
            zeroline=False,
            color='#2C3E50'
        ),
        yaxis=dict(
            title=f"<b>{ylabel}</b>",
            showgrid=True,
            gridcolor='#EAEAEA',
            zeroline=False,
            color='#2C3E50'
        ),
        legend=dict(
            bgcolor='rgba(255, 255, 255, 0.8)',
            bordercolor='#EAEAEA',
            borderwidth=1
        ) if show_legend else None,
```

```

showlegend=show_legend,
margin=dict(l=60, r=40, t=80, b=60),
font=dict(family="Arial, sans-serif", size=12, color="#2C3E50")
)
return fig

```

Question 1: How has global life expectancy evolved, and where were the shocks?

In [24]:

```

# Aggregate global average by year
trend = life.groupby("year")["life_expectancy"].mean().reset_index()

# Generate the line plot
fig = px.line(
    trend,
    x="year",
    y="life_expectancy",
    markers = True,
    color_discrete_sequence=["#16A085"]
)

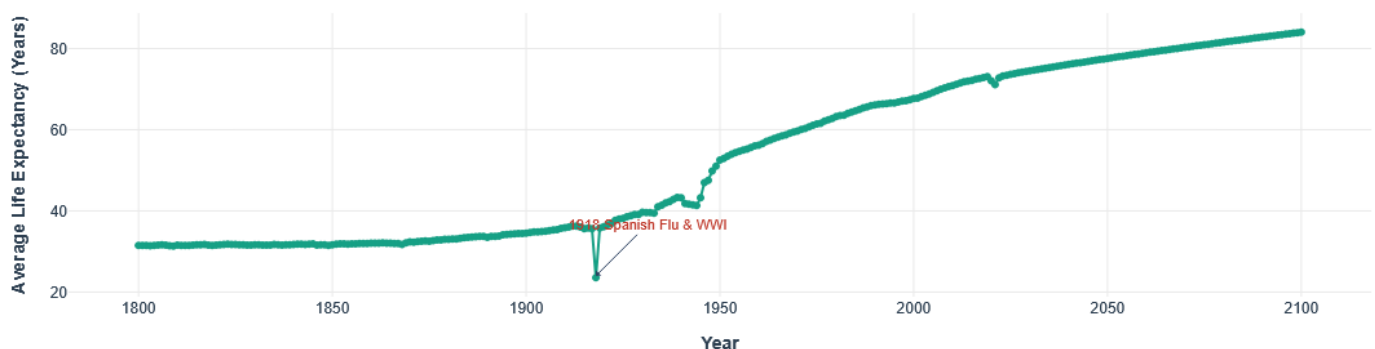
# Style using helper
fig = style_figure(
    fig,
    title="Global Average Life Expectancy (1800 - Present)",
    xlabel="Year",
    ylabel="Average Life Expectancy (Years)"
)

# Annotate the massive dip caused by the 1918 Spanish Flu
fig.add_annotation(
    x=1918, y=24,
    text="1918 Spanish Flu & WWI",
    showarrow=True,
    arrowhead=1,
    ax=40, ay=-40,
    font=dict(color="#C0392B", size=11)
)

fig.show()

```

Global Average Life Expectancy (1800 - Present)



Insights:

- A gradual increase from the early 1800s to the early 1900s, reflecting slow improvements in sanitation, nutrition, and public health.
- A much steeper rise after the mid-20th century, driven by advances such as antibiotics, vaccines, improved maternal and child healthcare, and expanded access to medical services.
- Stabilization in recent years, with gains becoming more gradual as many countries approach higher life expectancy levels.

Question 2: Which countries have experienced the most radical change?

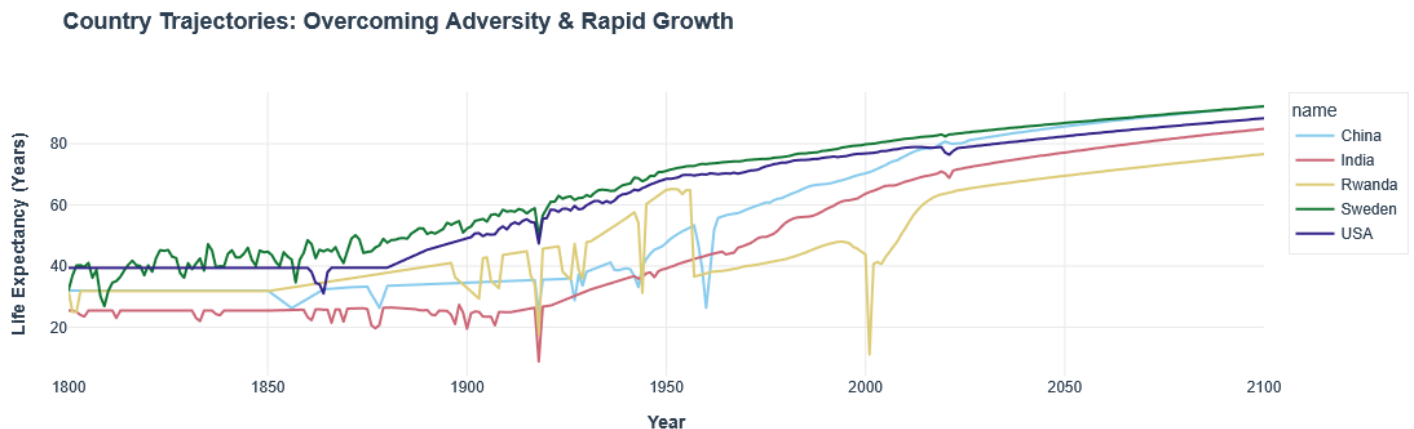
In [23]:

```
# Filter down to a few contrasting, illustrative countries
sample_countries = ["Sweden", "China", "USA", "Rwanda", "India"]
df_samples = life[life["name"].isin(sample_countries)]

fig_countries = px.line(
    df_samples,
    x="year",
    y="life_expectancy",
    color="name",
    color_discrete_sequence=px.colors.qualitative.Safe
)

# Style using helper
fig_countries = style_figure(
    fig_countries,
    title="Country Trajectories: Overcoming Adversity & Rapid Growth",
    xlabel="Year",
    ylabel="Life Expectancy (Years)"
)

fig_countries.show()
```



Insight

Rapid Convergence: Since the mid-1900s, developing nations like India and China have significantly closed the life expectancy gap with early developers like Sweden and the US.

Global life expectancy has increased substantially over time, demonstrating long-term improvements in healthcare, disease prevention, living standards, and socioeconomic development. The most rapid gains occur during the second half of the 20th century, while future projections indicate continued but slower improvements as populations approach biological and public health limits.

Question 3: How has the global distribution of lifespan shifted?

In [25]:

```
# Select target milestone years
milestones = [1820, 1870, 1920, 1970, 2020]
df_milestones = life[life["year"].isin(milestones)].copy()
df_milestones["year"] = df_milestones["year"].astype(str) # category conversion for plotting

fig_dist = px.box(
    df_milestones,
```

```

x="year",
y="life_expectancy",
points="outliers",
color="year",
)

fig_dist = style_figure(
    fig_dist,
    title="Global Lifespan Distribution Shift (50-Year Milestones)",
    xlabel="Milestone Year",
    ylabel="Life Expectancy (Years)",
    show_legend=False
)

fig_dist.show()

```



Insight

Inequality is Shrinking but Persists: The boxplot milestone visualization reveals a tight global lifespan range in 1820 (where almost everyone died young), which widened aggressively during industrialization, and is now compressing again near the top of the scale (as modern medicine and sanitation scale globally).

Question 4: How does Nigeria compare with neighboring countries?

In [26]:

```

neighbors = [
    "Nigeria",
    "Benin",
    "Cameroon",
    "Chad",
    "Niger",
]

neighbor_df = life[
    life["name"].isin(neighbors)
]

```

In [27]:

```

fig = px.line(
    neighbor_df,
    x="year",
    y="life_expectancy",
    color="name",
)

style_figure(
    fig,
    title="Nigeria vs Neighboring Countries: Life Expectancy (1800-2100)",
    xlabel="Year",

```

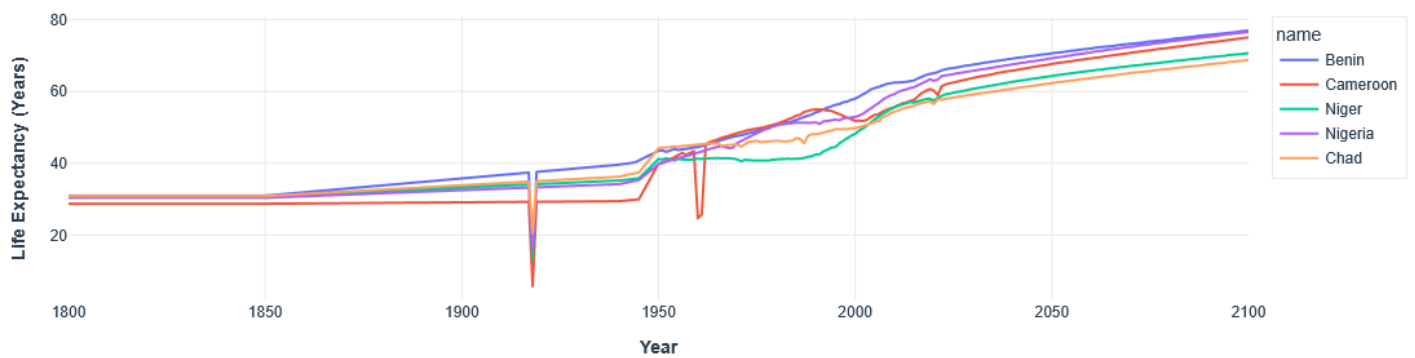
```

        ylabel="Life Expectancy (Years)"
    )

fig.show()

```

Nigeria vs Neighboring Countries: Life Expectancy (1800–2100)



In [28]:

```

avg_life = (
    neighbor_df
    .groupby("name", as_index=False) ["life_expectancy"]
    .mean()
    .sort_values("life_expectancy")
)

avg_life

```

Out[28]:

	name	life_expectancy
3	Niger	44.290000
1	Cameroon	44.682924
2	Chad	44.748472
4	Nigeria	46.594352
0	Benin	48.594518

In [29]:

```

fig = px.bar(
    avg_life,
    x="life_expectancy",
    y="name",
    orientation="h",
    color="life_expectancy",
    text_auto=".1f"
)

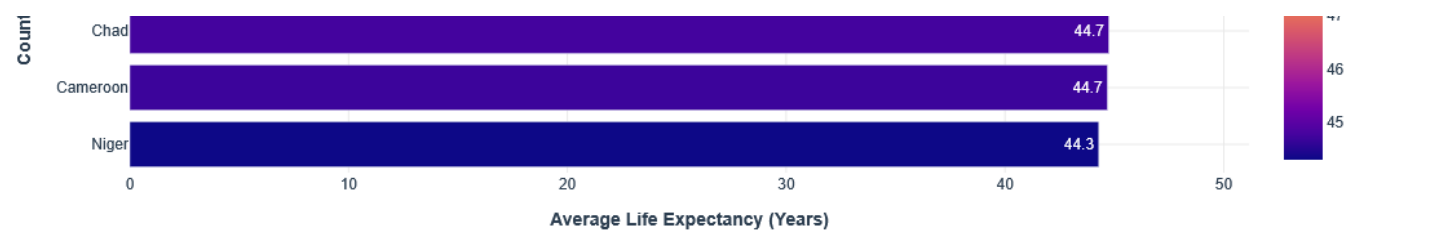
style_figure(
    fig,
    title="Average Life Expectancy: Nigeria vs Neighboring Countries",
    xlabel="Average Life Expectancy (Years)",
    ylabel="Country"
)

fig.show()

```

Average Life Expectancy: Nigeria vs Neighboring Countries





In [30]:

```
comparison = (
    neighbor_df
    .pivot(
        index="name",
        columns="year",
        values="life_expectancy"
    )
)

comparison["Improvement"] = (
    comparison[2024] - comparison[1800]
)

comparison = (
    comparison["Improvement"]
    .sort_values(ascending=False)
    .reset_index()
)

comparison
```

Out[30]:

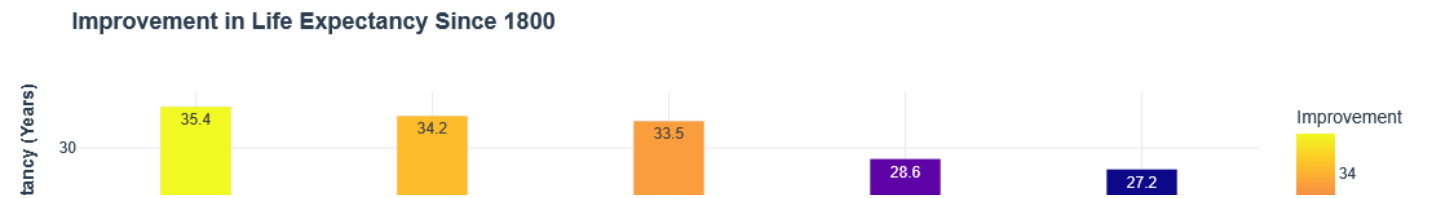
	name	Improvement
0	Benin	35.42
1	Nigeria	34.22
2	Cameroon	33.54
3	Niger	28.59
4	Chad	27.21

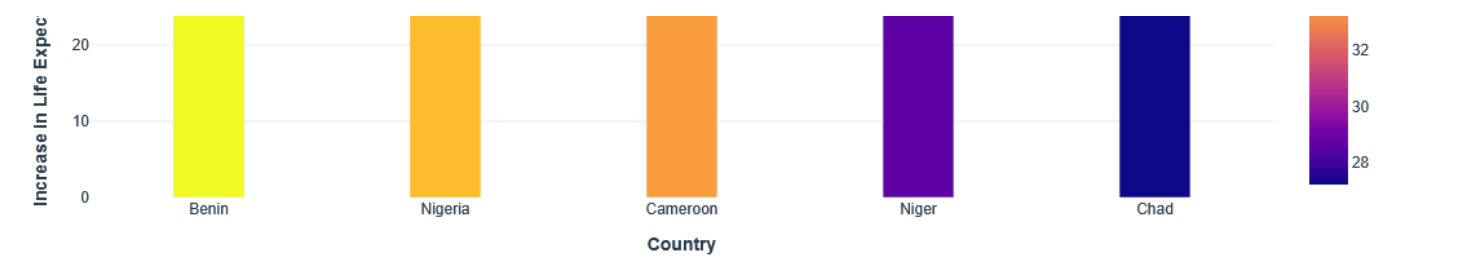
In [31]:

```
fig = px.bar(
    comparison,
    x="name",
    y="Improvement",
    color="Improvement",
    text_auto=".1f"
)
# Make the bars thinner
fig.update_traces(width=0.3)

style_figure(
    fig,
    title="Improvement in Life Expectancy Since 1800",
    xlabel="Country",
    ylabel="Increase in Life Expectancy (Years)"
)

fig.show()
```





Insight:

Nigeria has experienced substantial gains in life expectancy over the past two centuries, reflecting improvements in healthcare, vaccination programs, sanitation, and living conditions. However, when compared with neighboring countries such as Benin, Cameroon, Chad, and Niger, its relative position depends on the time period examined. The trend analysis highlights whether Nigeria has kept pace with regional progress or whether disparities in health outcomes remain.

In []:

Question 5: What are the countries with greatest improvement?

In [35]:

```
change = (\n    life\n    .pivot(index="name",\n            columns="year",\n            values="life_expectancy")\n)\n\nchange["Increase"] = (\n    change[2026]-change[1800]\n)\n\nchange = (\n    change["Increase"]\n    .sort_values()\n    .reset_index()\n    .dropna()\n)
```

In [36]:

```
change.tail(10)
```

Out[36]:

	name	Increase
174	Kyrgyz Republic	51.68
175	Nicaragua	51.81
176	Israel	51.81
177	Iran	53.71
178	Italy	53.81
179	Spain	54.10
180	Malta	54.80
181	Kuwait	55.71
182	Singapore	56.71
183	South Korea	57.90

In [38]:

```
fig = px.bar(
    change.tail(10),
    x="Increase",
    y="name",
    orientation="h",
    color="Increase",
    color_continuous_scale="Viridis",
    text_auto=".1f"
)

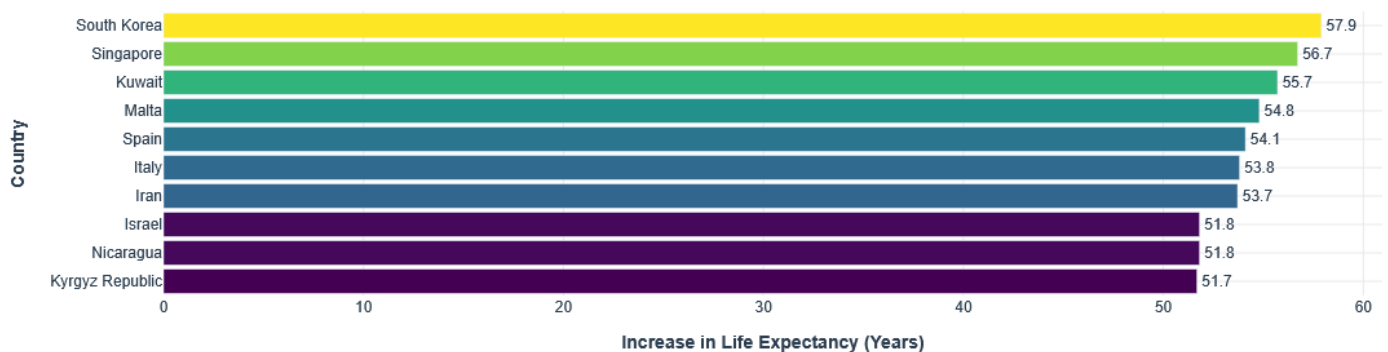
fig.update_traces(
    textposition="outside",
    hovertemplate="<b>{y}</b><br>Increase: {x:.1f} years<extra></extra>"
)

fig.update_layout(
    bargap=0.15,
    coloraxis_showscale=False
)

style_figure(
    fig,
    title="Top 10 Countries with the Greatest Improvement in Life Expectancy (1800–2026)"
,
    xlabel="Increase in Life Expectancy (Years)",
    ylabel="Country"
)

fig.show()
```

Top 10 Countries with the Greatest Improvement in Life Expectancy (1800–2026)



Question 6: What will be the life expectancy in Africa in the future?

In [39]:

```
import pycountry
import pycountry_convert as pc
```

In [40]:

```
def get_continent(country_name):
    """
    Return the continent for a country name.
    Returns None if the country cannot be matched.
    """
    try:
        country = pycountry.countries.lookup(country_name)

        continent_code = pc.country_alpha2_to_continent_code(
            country.alpha_2
        )

        continent_name = {
```

```

        "AF": "Africa",
        "AS": "Asia",
        "EU": "Europe",
        "NA": "North America",
        "OC": "Oceania",
        "SA": "South America",
    }

    return continent_name.get(continent_code)

except Exception:
    return None

```

In [41]:

```

life["Continent"] = life["name"].apply(get_continent)

life.head()

```

Out[41]:

	geo	name	year	life_expectancy	Continent
0	afg	Afghanistan	1800	28.21	Asia
1	ago	Angola	1800	26.98	Africa
2	alb	Albania	1800	35.40	Europe
3	are	UAE	1800	30.70	None
4	arg	Argentina	1800	33.20	South America

In [48]:

```

africa = life[life["Continent"] == "Africa"]

africa_avg = (
    africa.groupby("year", as_index=False)["life_expectancy"]
    .mean()
)

```

In [49]:

```

fig = go.Figure()

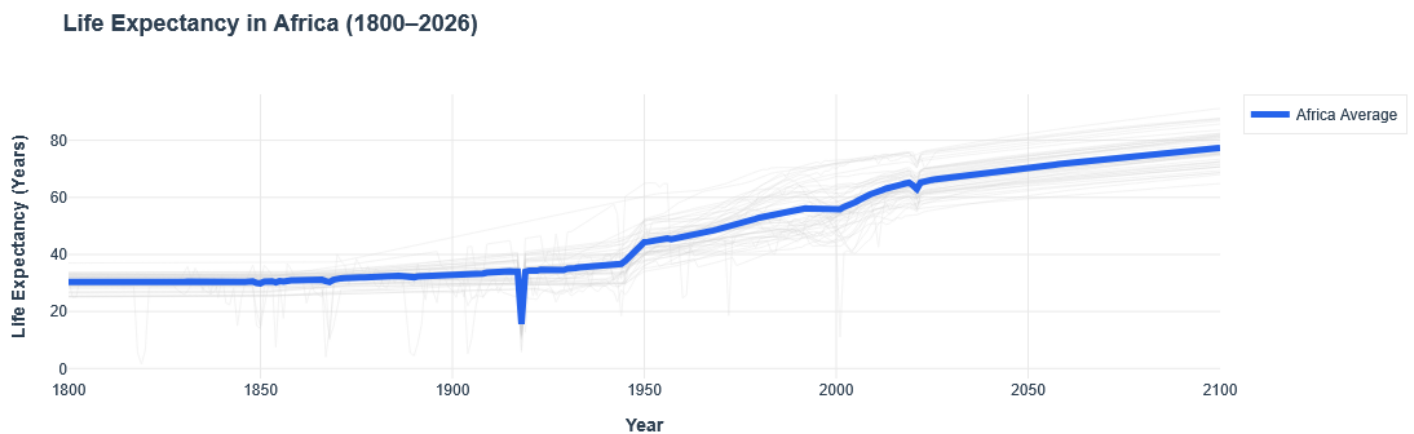
# Individual countries
for country, df in africa.groupby("name"):
    fig.add_trace(
        go.Scatter(
            x=df["year"],
            y=df["life_expectancy"],
            mode="lines",
            line=dict(width=1, color="lightgray"),
            opacity=0.25,
            showlegend=False,
            hoverinfo="skip"
        )
    )

# Continental average
fig.add_trace(
    go.Scatter(
        x=africa_avg["year"],
        y=africa_avg["life_expectancy"],
        mode="lines",
        name="Africa Average",
        line=dict(color="#2563EB", width=5)
    )
)

fig = style_figure(

```

```
fig,
"Life Expectancy in Africa (1800–2026)",
"Year",
"Life Expectancy (Years)"
)
fig.show()
```



Insight:

- Average life expectancy has increased dramatically over the last two centuries.
- By 2026, it is approximately **65–70** years.
- The long-term trend suggests continued improvement, although future gains will depend on healthcare investments, disease control, political stability, and socioeconomic development.

Key Insights

1. Life expectancy has increased significantly over time.

- The global average life expectancy has risen steadily from the early 1800s to 2026.
- This reflects improvements in medicine, sanitation, nutrition, education, and public health.

1. Africa has experienced remarkable progress.

- Although Africa started with one of the lowest average life expectancies, the continent has shown consistent improvement over the last century.
- The sharpest increase occurred after the 1950s due to advances in healthcare and disease prevention.

1. Regional disparities still exist.

- Europe, North America, and Oceania generally maintain higher life expectancies than Africa and parts of Asia.
- This suggests that socioeconomic development and healthcare accessibility remain important determinants of longevity.

1. Some countries improved faster than others.

- Countries that invested heavily in healthcare infrastructure, education, and vaccination programs experienced larger gains in life expectancy.
- Political instability, conflict, and disease outbreaks slowed progress in some regions.

1. Future projections remain optimistic.

- Based on historical trends, Africa's average life expectancy is expected to continue increasing.
- However, the rate of improvement may gradually slow as countries approach higher life expectancy levels.

Project Summary

This project analyzed over two centuries of life expectancy data across countries and continents to understand how longevity has changed over time.

human longevity has changed over time.

The analysis involved:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Trend analysis
- Continental comparisons
- Country-level comparisons
- Interactive visualizations using Plotly

The findings demonstrate that global life expectancy has improved dramatically due to continuous advances in healthcare, technology, education, and living standards. Despite this progress, inequalities remain across different regions, emphasizing the need for sustained investments in healthcare systems and socioeconomic development.

Conclusion

The analysis demonstrates one of humanity's greatest achievements: people are living significantly longer than they did two centuries ago.

Although challenges remain, particularly in developing regions, the overall trajectory is positive. Continued investments in healthcare, education, sanitation, and economic development will likely lead to further increases in life expectancy across the world.